2.4 Differentiating the Product and the Quotient of Two Functions; Higher-Order Derivatives

Name _____ Period ____ Date _____

AP Calculus 2.4 · Student sheet

I can…

- use the Product Rule.
- use the Quotient Rule and the Power Rule for negative exponents.
- find second and higher-order derivatives, including acceleration.

Vocabulary · fill in the missing word

Product Rule — $(fg)' =$ _____

Quotient Rule — $\left(\frac{f}{g}\right)' =$ _____, $g \neq 0$

Second derivative — _____

Part 1 · Is $(fg)' = f'g'$?

$f(x) = x$ and $g(x) = x^2$, so $f(x)g(x) = x^3$.

Guess	Result	Equal to $3x^2$?
$f'(x)g'(x)$		
$f'(x)g(x) + f(x)g'(x)$		

Part 2 · Notes and examples (fill in each note, then work the example)

Product Rule: $(fg)' = f'g +$ _____.

Example 1

Find $f'(x)$ for $f(x) = (x^2 + 1)(3x - 2)$.

Example 2

Find $f'(x)$ for $f(x) = xe^x$.

Quotient Rule: $(f/g)' = (f'g - fg')/$_____.

Example 3

Find $f'(x)$ for $f(x) = \frac{x^2 + 1}{2x - 1}$.

Power Rule, negative exponents: $d/dx\ x^{-n} =$ _____.

Example 4

Find each derivative.

a. $y = \frac{1}{x^3}$

b. $y = \frac{5}{x^2}$

Higher-order derivatives: f'' is the derivative of f'; if $s(t)$ is position, $s''(t)$ is the _____.

Example 5

a. $f(x) = 3x^4 - 2x^3 + x$. Find $f''(x)$.

b. $s(t) = t^3 - 6t^2$. Find the acceleration at $t = 3$.

Part 3 · Partner practice

Partner 1

Find $f'(x)$ for $f(x) = \frac{3x - 2}{x + 4}$.

Partner 2

(AP® Practice) If $f(x) = x^3 + 4x^2 - 2x$, then $f''(1) =$

(A) 12 (B) 14 (C) 9 (D) 6